



Koneru Lakshmaiah Education Foundation

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STUDENT CASE STUDY EXECUTION REPORT

Department	FED
Subject	DSA-2
Academic Year	I-III Semester (2025-26)
Faculty Name	Dr.Kavitha k
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Roll Number	2520040009
Branch	ECE
Course Outcome	CO3

Case Study Topic: Airline Connectivity Analysis using DSU Rollback (Set 9 Q3)

Work Done Summary:

In this case study, we implemented a Disjoint Set Union (DSU) with Rollback to analyse connectivity in an airline network. Airports were represented as vertices and flight routes as graph edges. Union-by-Rank was used to merge connected components efficiently. A rollback mechanism was implemented to restore previous graph states after processing each query. Connectivity results were compared before and after the removal of routes. The implementation demonstrated the effectiveness of DSU Rollback for dynamic connectivity problems in transportation networks.

Implementation Details:

1. Implemented parent and rank arrays.
2. Applied union-by-rank.
3. Stored rollback history.
4. Restored graph state after rollback.
5. Compared with BFS/DFS approach.

Result: Screenshots / Output

```
Microsoft Windows [Version 10.0.26200.8524]
(c) Microsoft Corporation. All rights reserved.

C:\Users\netta\Music\DSA project>javac AirlineConnectivityBFS.java

C:\Users\netta\Music\DSA project>java AirlineConnectivityBFS
AIRLINE CONNECTIVITY ANALYSIS

Query 1:
If BLR-BOM is removed, can BLR reach DEL?
Answer: YES

Query 2:
If MAA-COK is removed, can BLR reach COK?
Answer: NO

Query 3:
If BOM-DEL and AMD-DEL are removed,
can DEL still be reached?
Answer: YES

C:\Users\netta\Music\DSA project>|
```

Connected (0,3): true

After Rollback Connected

(0,3): false **Time**

Complexity:

Find: $O(\log n)$

Union: $O(\log n)$

Rollback: $O(1)$

GITHUB LINK : <https://github.com/pranay-0508/DSA---CASE-STUDIES-3.git>

Student Signature	Faculty Signature